ECO 331

Economic History

MALTHUS, POPULATION, HUMAN CAPITAL

Jonathan Conning
Spring 2026

Readings

- OG 2 "Lost in Stagnation" 27-41
- OG 3 "The Storm Beneath the Surface", 43-55.
- (optional: KR 5 Fewer babies?)

Student presentation:

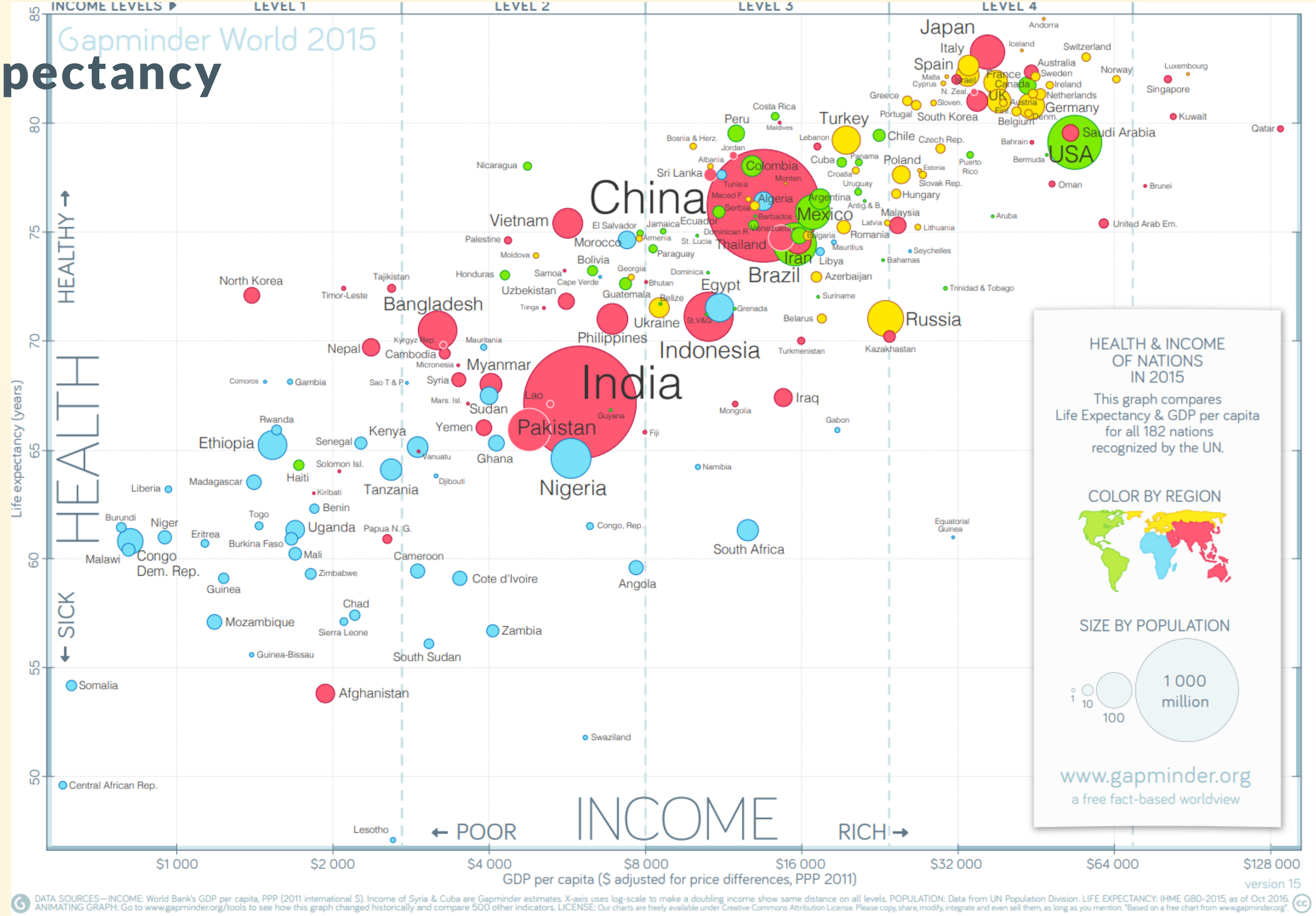
- Nunn, Nathan, and Nancy Qian. 2011. “The Potato’s Contribution to Population and Urbanization: Evidence from a Historical Experiment.” *The Quarterly Journal of Economics* 126 (2): 593–650. (PDF)

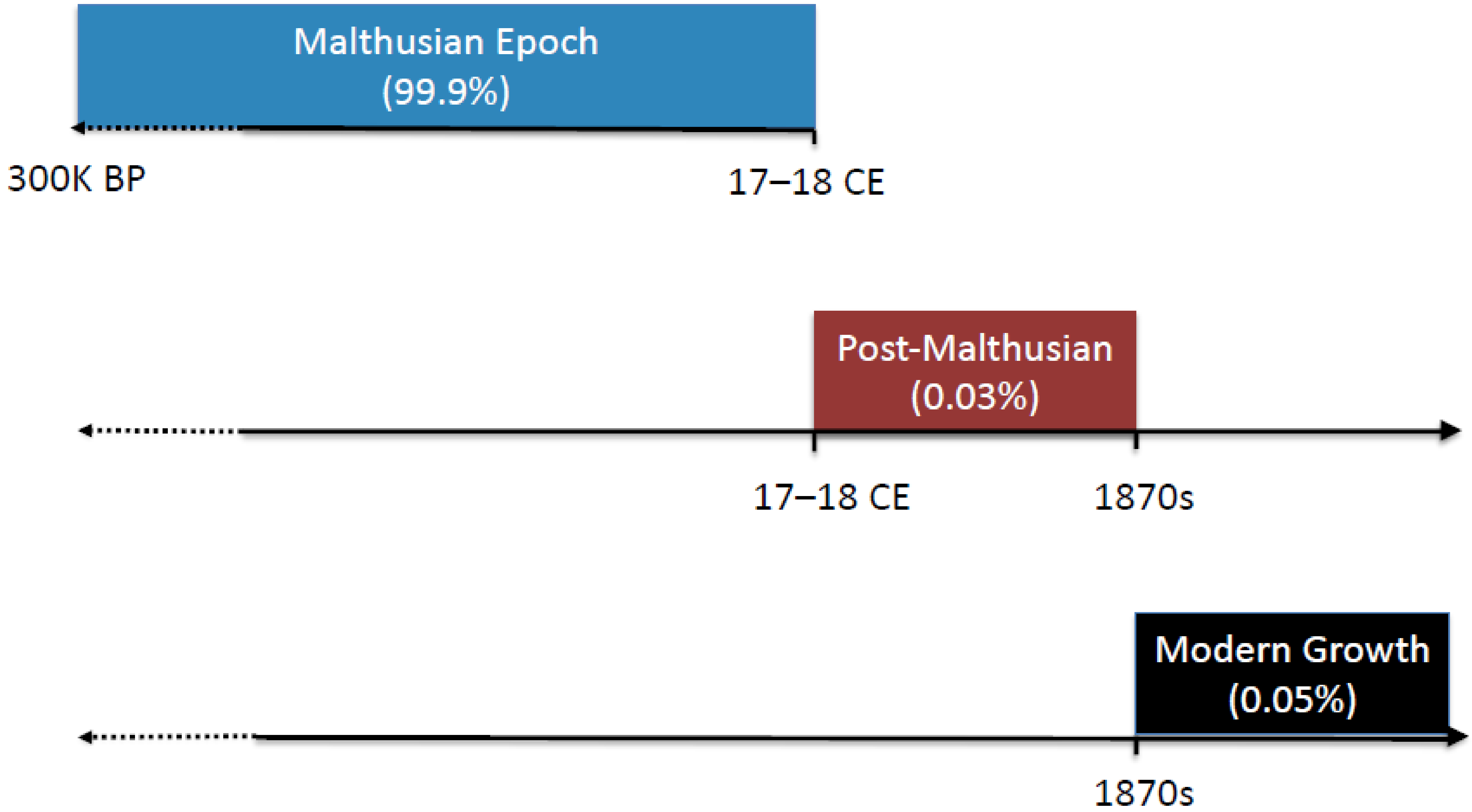
Demography in the pre-industrial world

- Life expectancy in Roman Egypt ~27-28 years (Harper, 2017)
    - LE in poorest countries today generally higher than pre-industrial past.
- Infant mortality was high (Harper, 2017)
    - 30% of babies died <1 yr in Roman world
- Life expectancy low even in richest parts of the world
    - In 1700 England it was 38 years at birth (considerably higher than France, or Roman Empire)
    - Queen Anne (rule 1702-1714): 17 pregnancies, 4 live births, no child survived past 10. She died at 49.

Life Expectancy

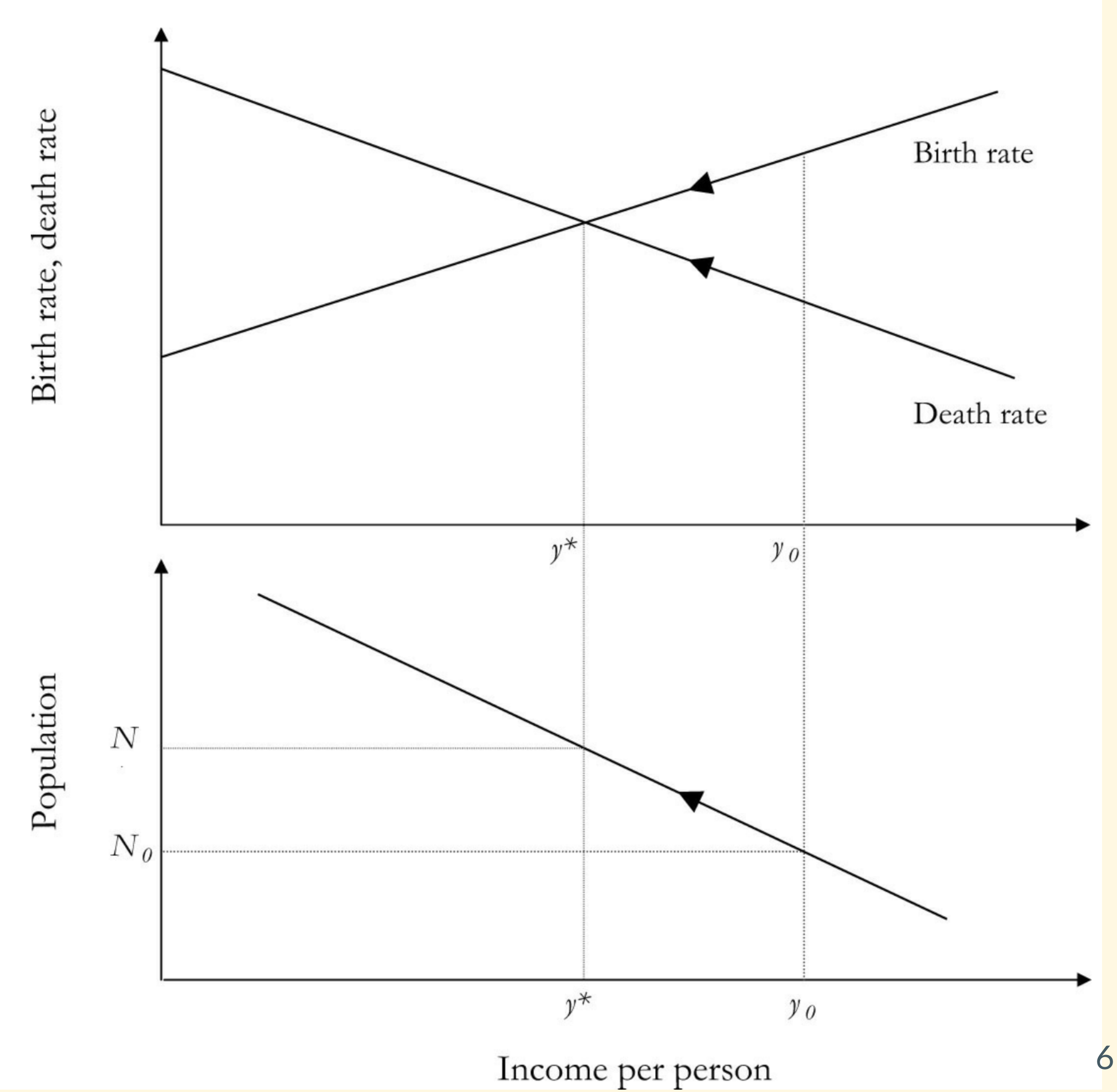
[link]()




Malthusian Epoch
(99.9%)
300K BP
17–18 CE
Post-Malthusian
(0.03%)
17–18 CE
1870s
Modern Growth
(0.05%)
1870s

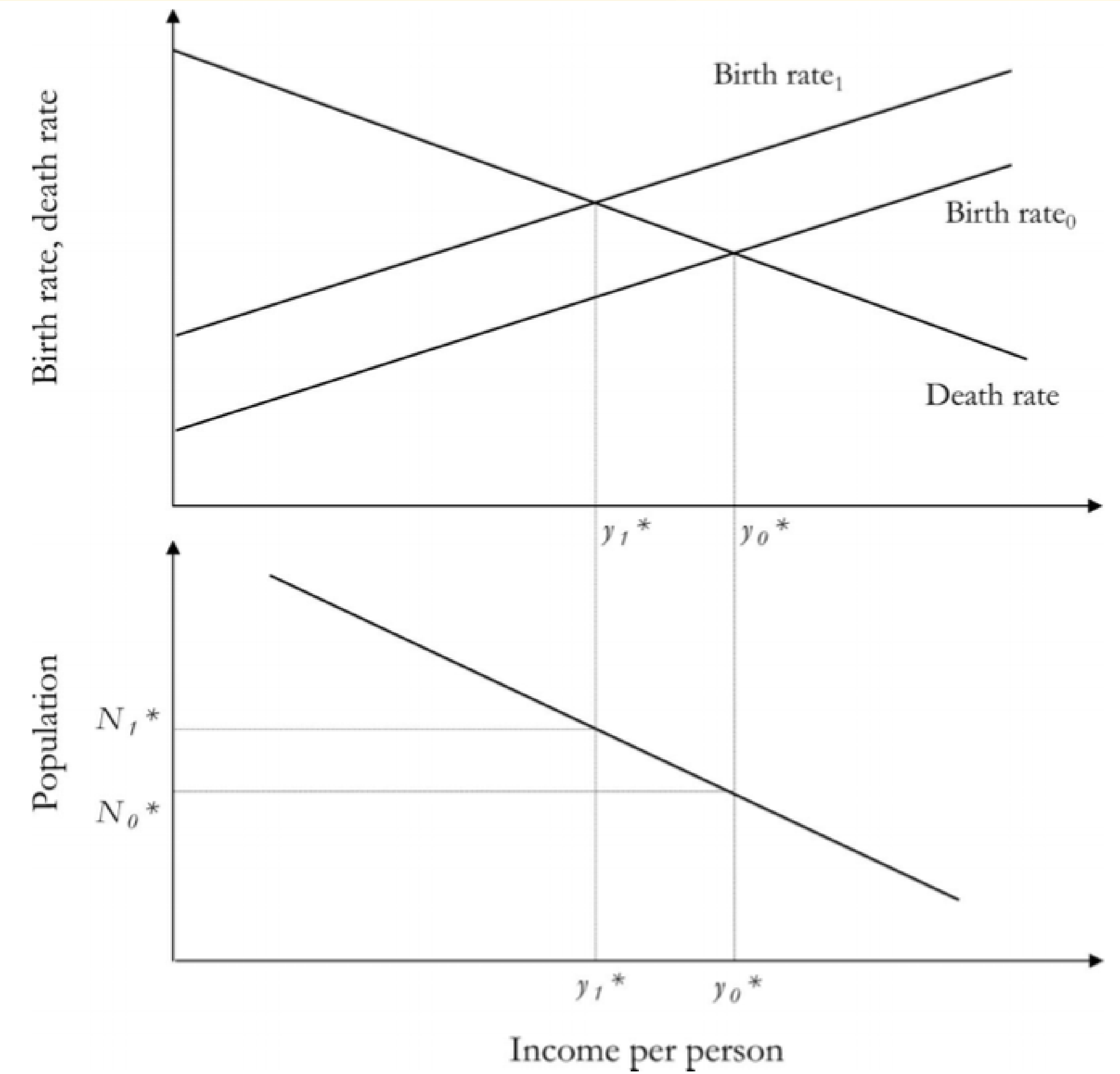
Malthus Model

- y income per capita
- TOP:
  - $b(y) \uparrow$ with y
  - $d(y) \downarrow$ with y
  - $b(y) > d(y) \rightarrow pop \uparrow$
- BOTTOM:
  - $y \downarrow$ as pop $N \uparrow$ (diminishing returns)
- IMPLICATION:
  - Always return to steady state:
  - pop N and income y^*



Effect of lower birth date

- Malthus' "preventative check"
- Efforts lower birth rate for any given y
- Examples
  - delay age of marriage
  - birth control
- At initial y_1^* deaths exceed births
  - popn falls N_1^* to N_0^*
  - income rises y_1^* to y_0^*
- New steady state.



Explicit model (optional)

A simple system of differential equations determine population N and total resources Y (and hence $y = Y/N$) at each time instant t.

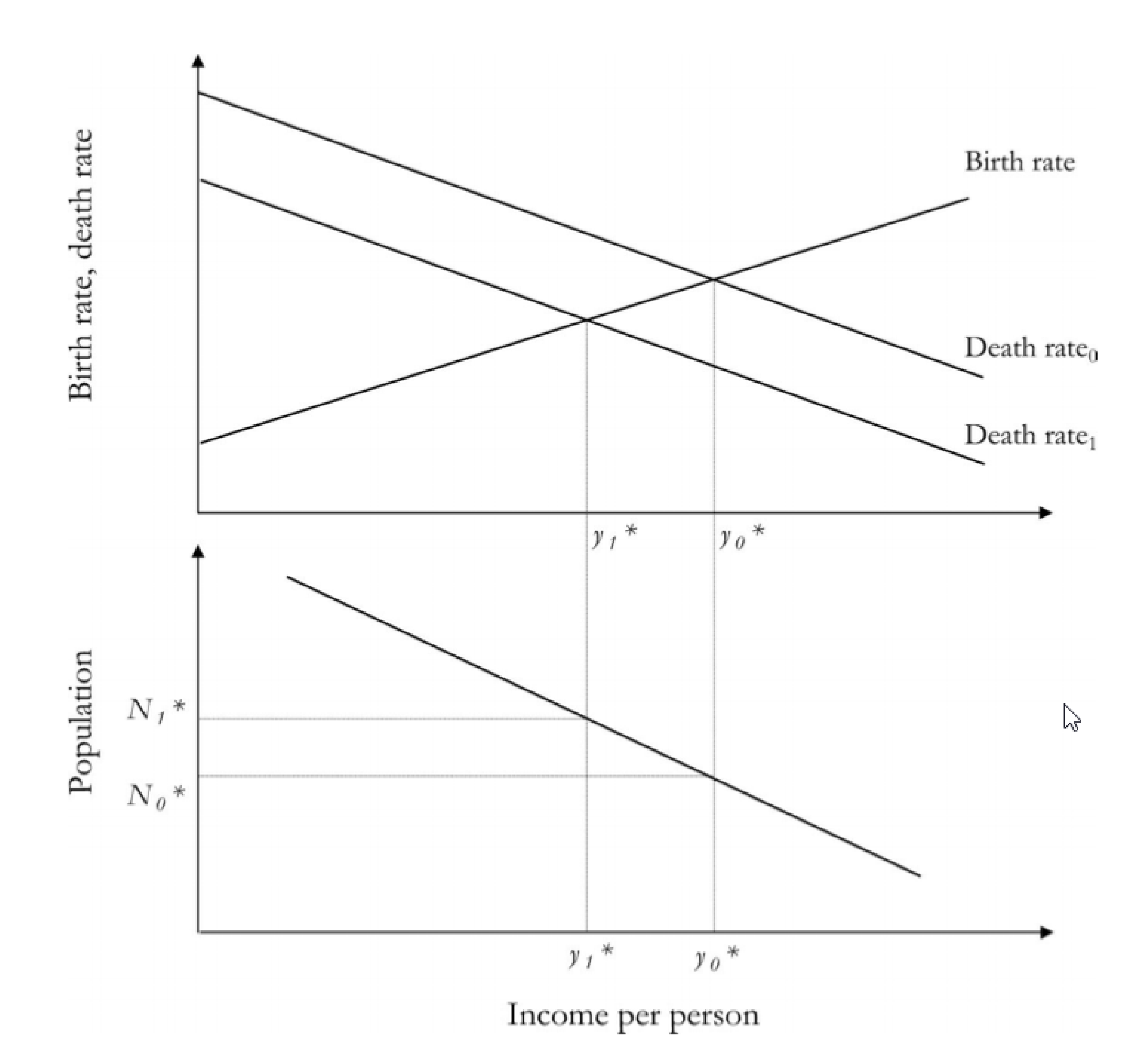
$$\frac{dN}{dt} = N(b(r) - d(r))$$
$$\frac{dY}{dt} = g - cN$$

- $y = \frac{Y}{N}$ (per capita resources)
- $b(y) = b_0 y$ (birth rate increases linearly with resources)
- $d(y) = d_0 e^{-\alpha y}$ (death rate decreases exponentially with resources)
- g : resource growth rate
- c : resource consumption rate per individual

Effect of raised death rate

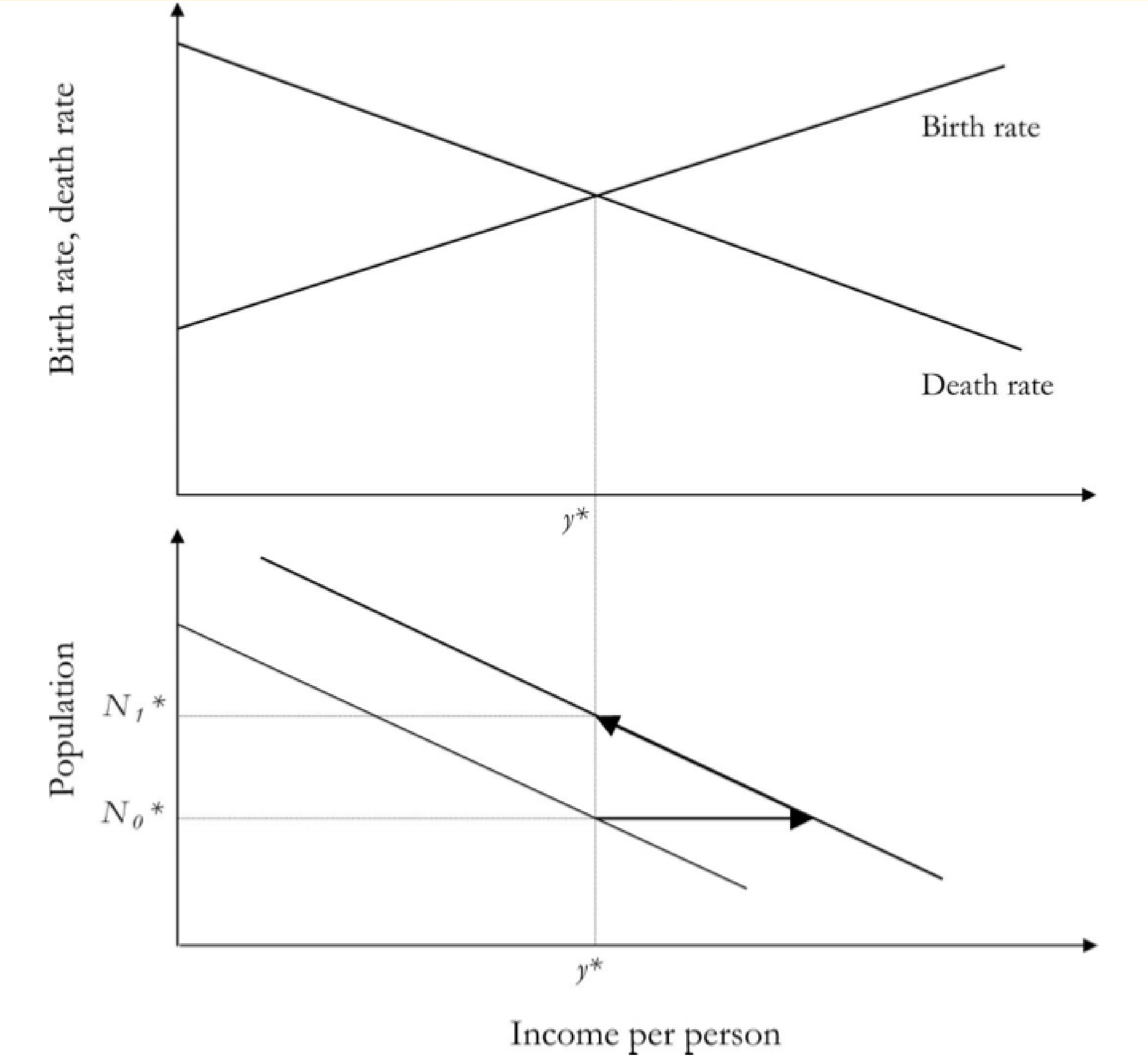
(from 1 to 0 in the diagram)

- e.g. epidemic
- At initial y_1^*, deaths now exceed births
  - popn falls N_1^* to N_0^*
  - income rises y_1^* to y_0^*
- New steady state.

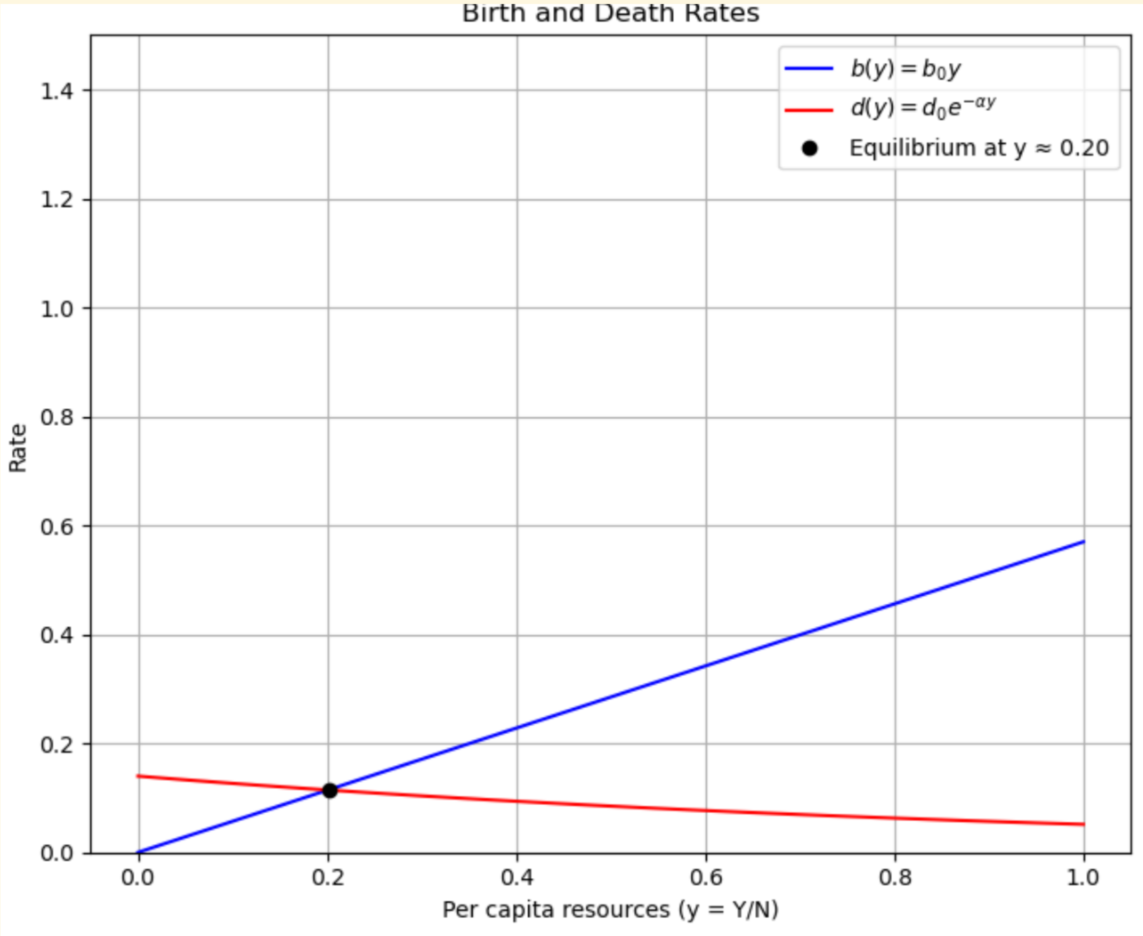


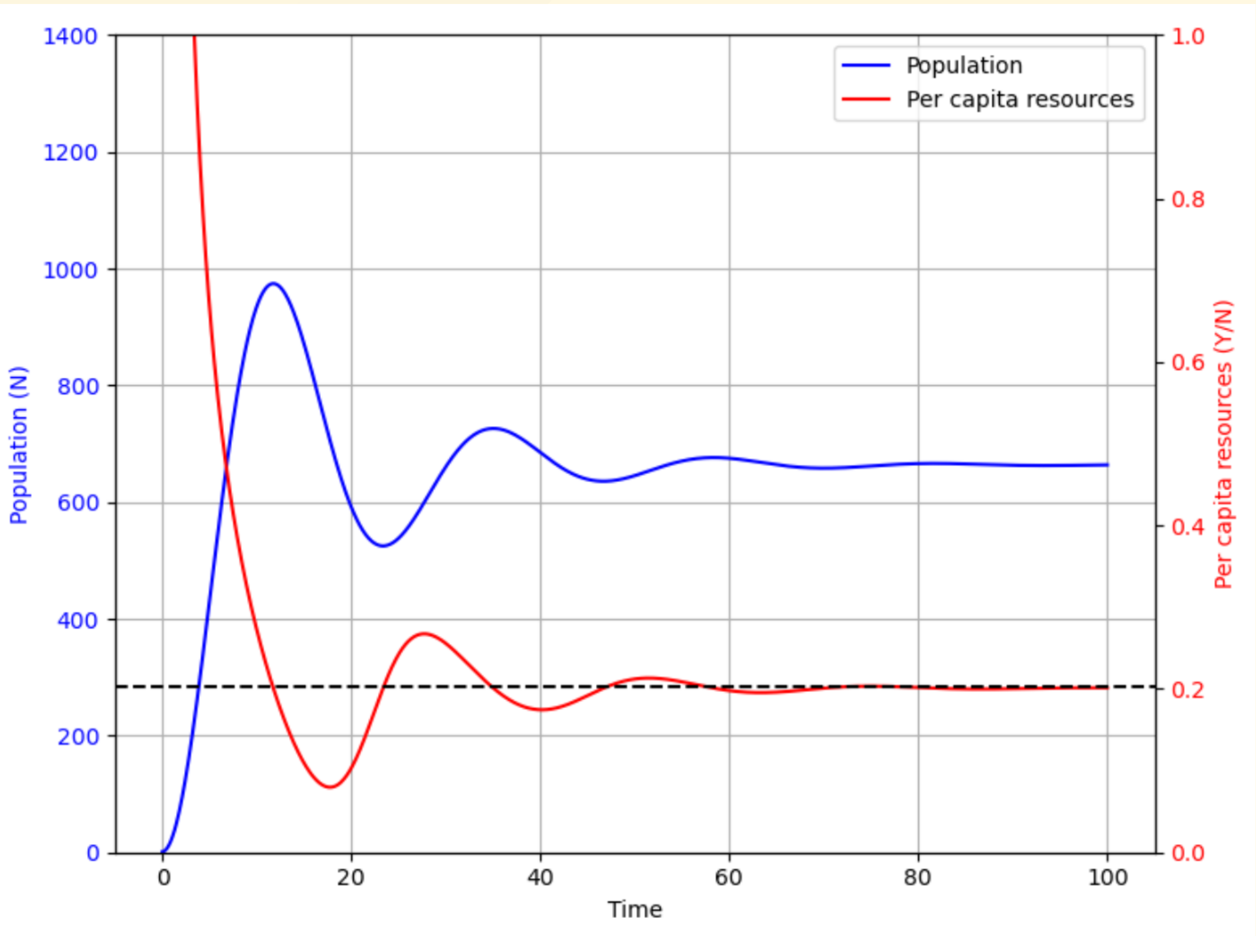
Effect of isolated Technological advance

- Starting from (y^*, N_0^*)
- bottom line shifts out (higher y at any given N)
- Births now exceed Deaths
  - popn rises to N_1^*
  - income returns to $y*$

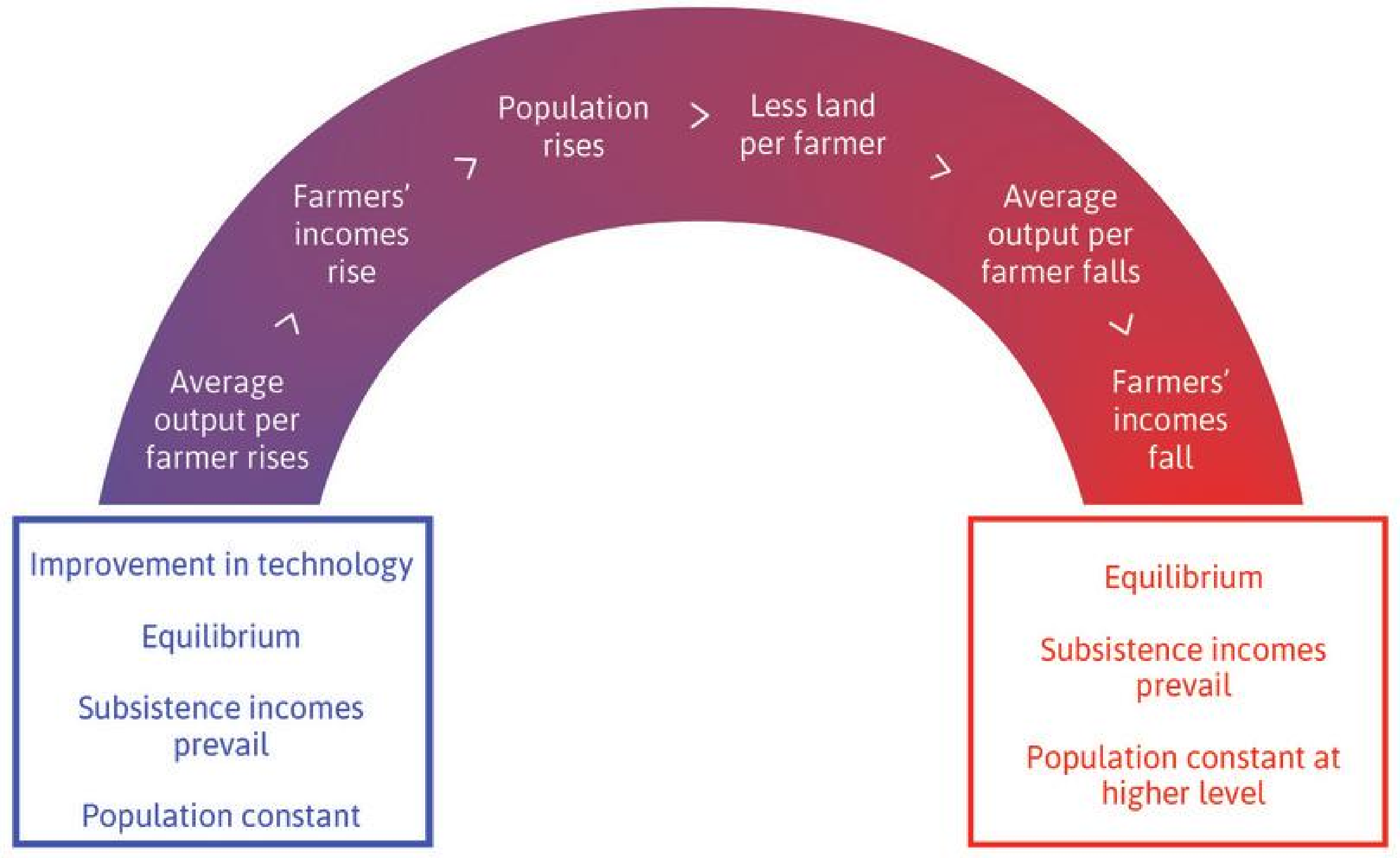


Malthusian Dynamics (simulation) [colab notebook link]





Improvement in Technology in Malthusian World



Earlier Onset of Agriculture

- Tech advance $\rightarrow$
- Popn $\rightarrow$ tech advance
  - more people, diversity, specialization, human capital.
- Earlier onset $\rightarrow$ better technology
  - disease immunity, soldiers, etc
  - Not higher incomes per capita
- Absorb, conquer, and/or exterminate/displace other societies

Galor, p3:

Most people of a few centuries ago led lives comparable to those of their remote ancestors – and most other individuals around the globe – millennia ago, rather than to those of their current descendants. The living conditions of an English farmer at the turn of the sixteenth century were similar to those of an eleventh-century Chinese serf, a Mayan peasant fifteen hundred years ago, a fourth-century BCE Greek herder, an Egyptian farmer 5,000 years ago, or a shepherd in Jericho 11,000 years ago. But since the dawn of the nineteenth century, a split second compared to the span of human existence, life expectancy has more than doubled, and per capita incomes have soared twenty-fold in the most developed regions of the world, and fourteen-fold on Planet Earth as a whole (Fig. 1).[2]

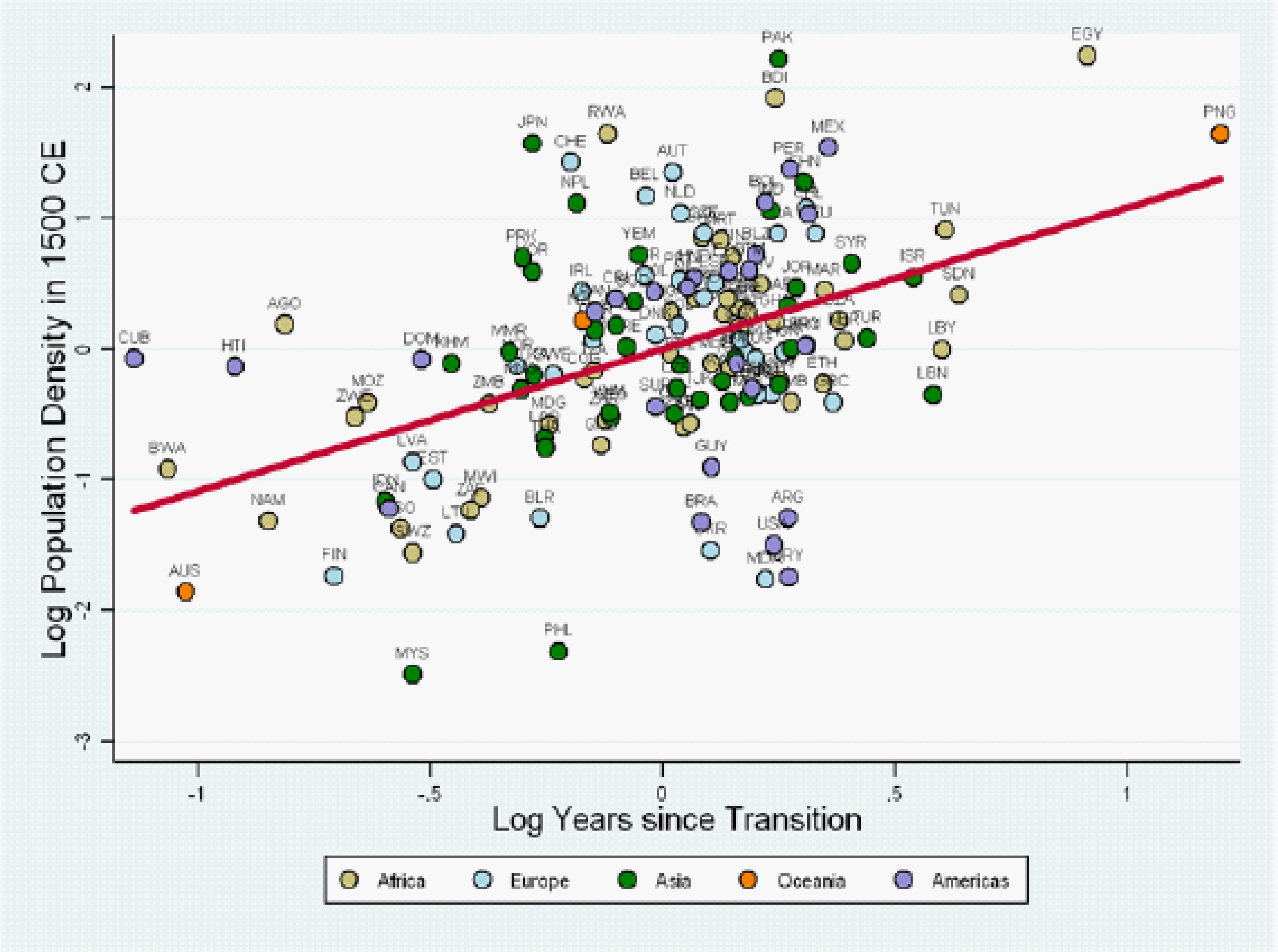
Pop Growth in Europe before 1800

- High birth rate
- High mortality
- Slow pop growth before 1800
- Table shows implied number of surviving children per woman:

Table 2.1 Populations in Western Europe, 1300 and 1800

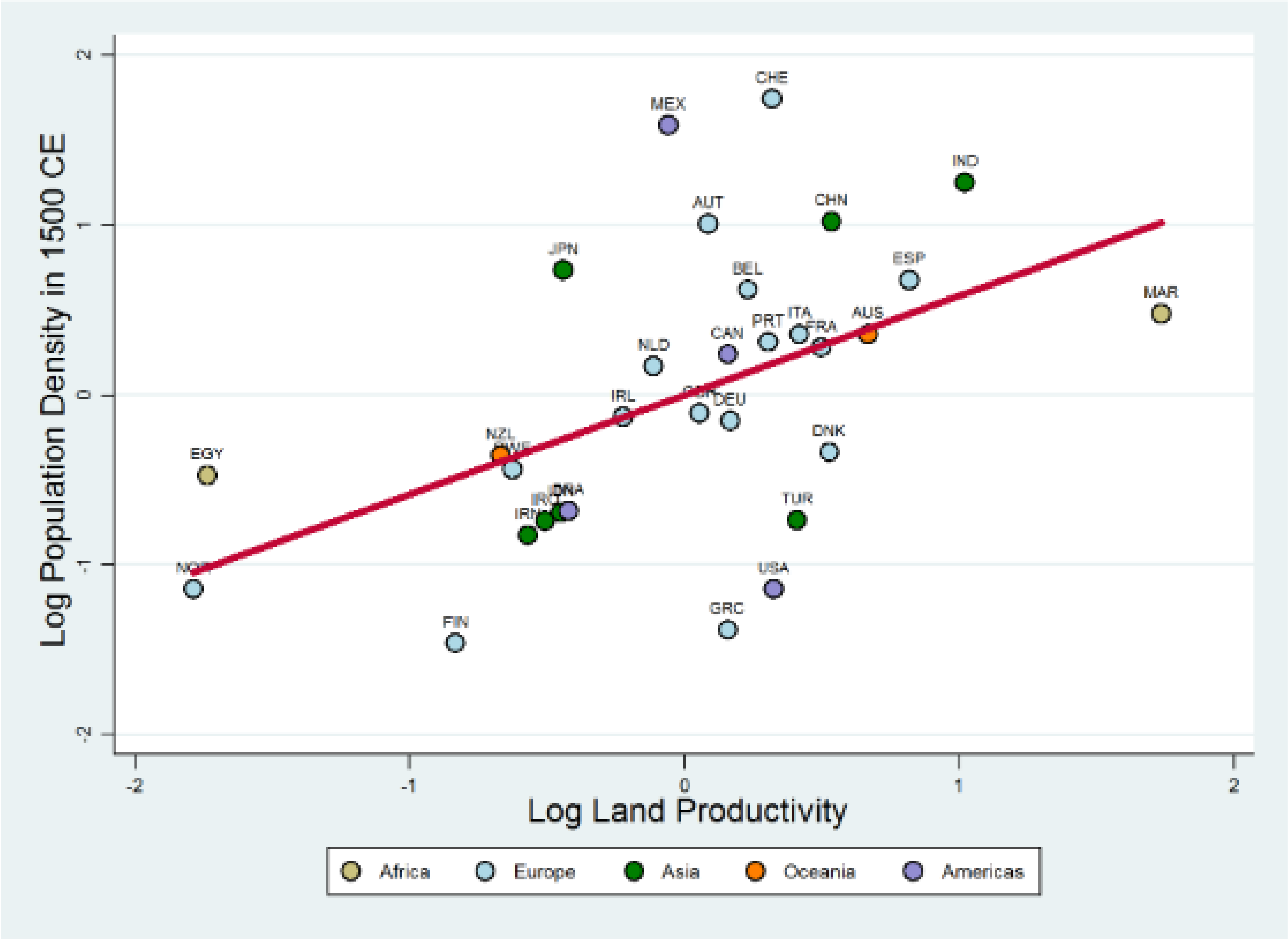
Location	Population ca. 1300	Population ca. 1800	Surviving children per woman
Norway[a]	0.40	0.88	2.095
Southern Italy[b]	4.75	7.9	2.061
France[c]	17.0	27.2	2.056
England[d]	5.8	8.7	2.049
Northern Italy[b]	7.75	10.2	2.033
Iceland[a]	0.084	0.047	1.930

Sources: [a]Tomasson, 1977, 406. [b]Federico and Malanima, 2004, table 4. [c]Le Roy Ladurie, 1981, 13.[d]Clark, 2007a, 120.

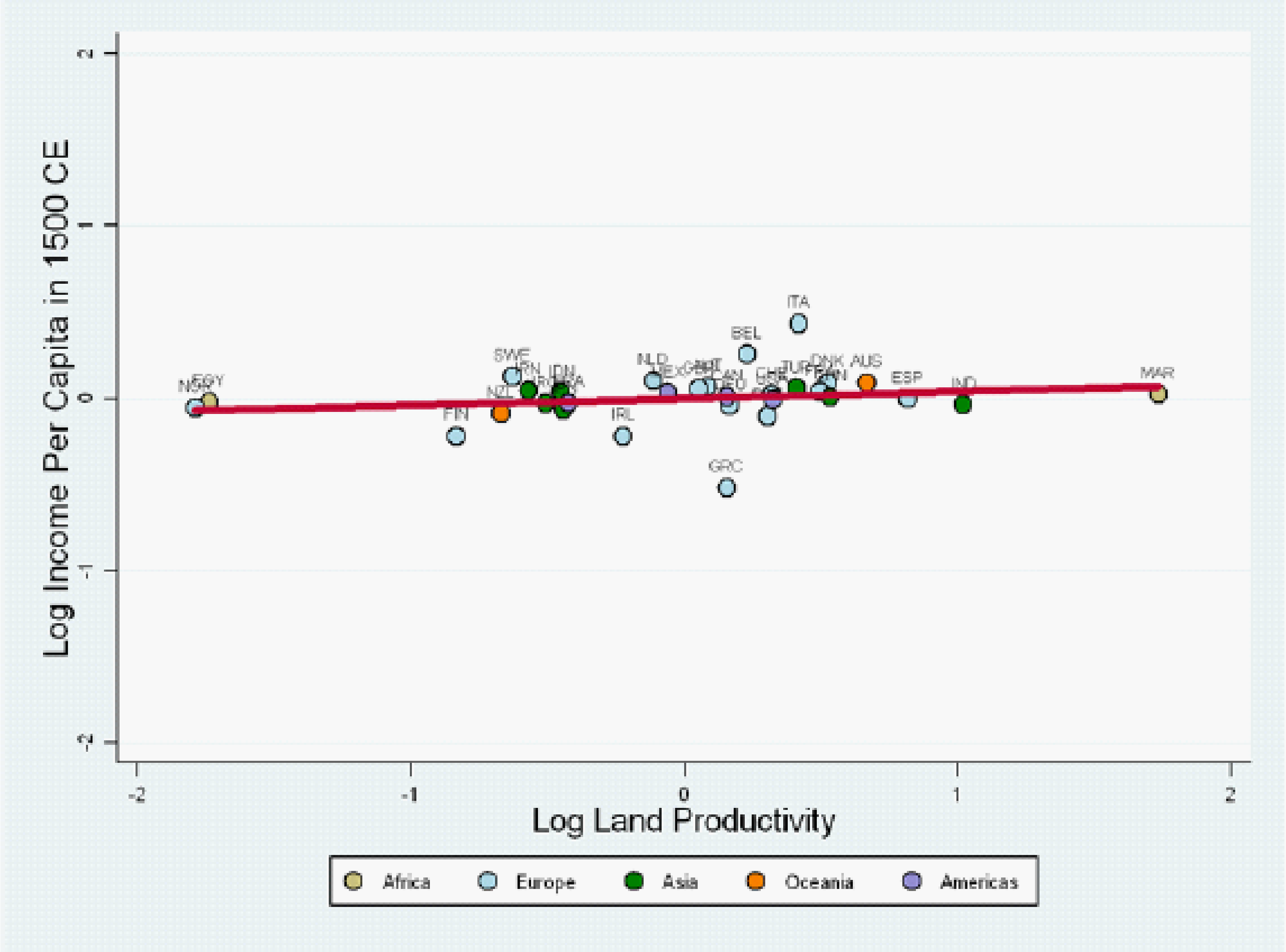


Years elapsed since the Neolithic Transition is a proxy for technological levels in 1500.

Conditional on land productivity, geographical factors, and continental fixed effects.



(b) The Partial Effect of Land Productivity on Population Density in 1500 CE



Conditional on transition timing, geographical factors, and continental fixed effects.

Source: Ashraf-Galor (AER 2011)

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Population through the ages video

Steady state need not mean minimum subsistence

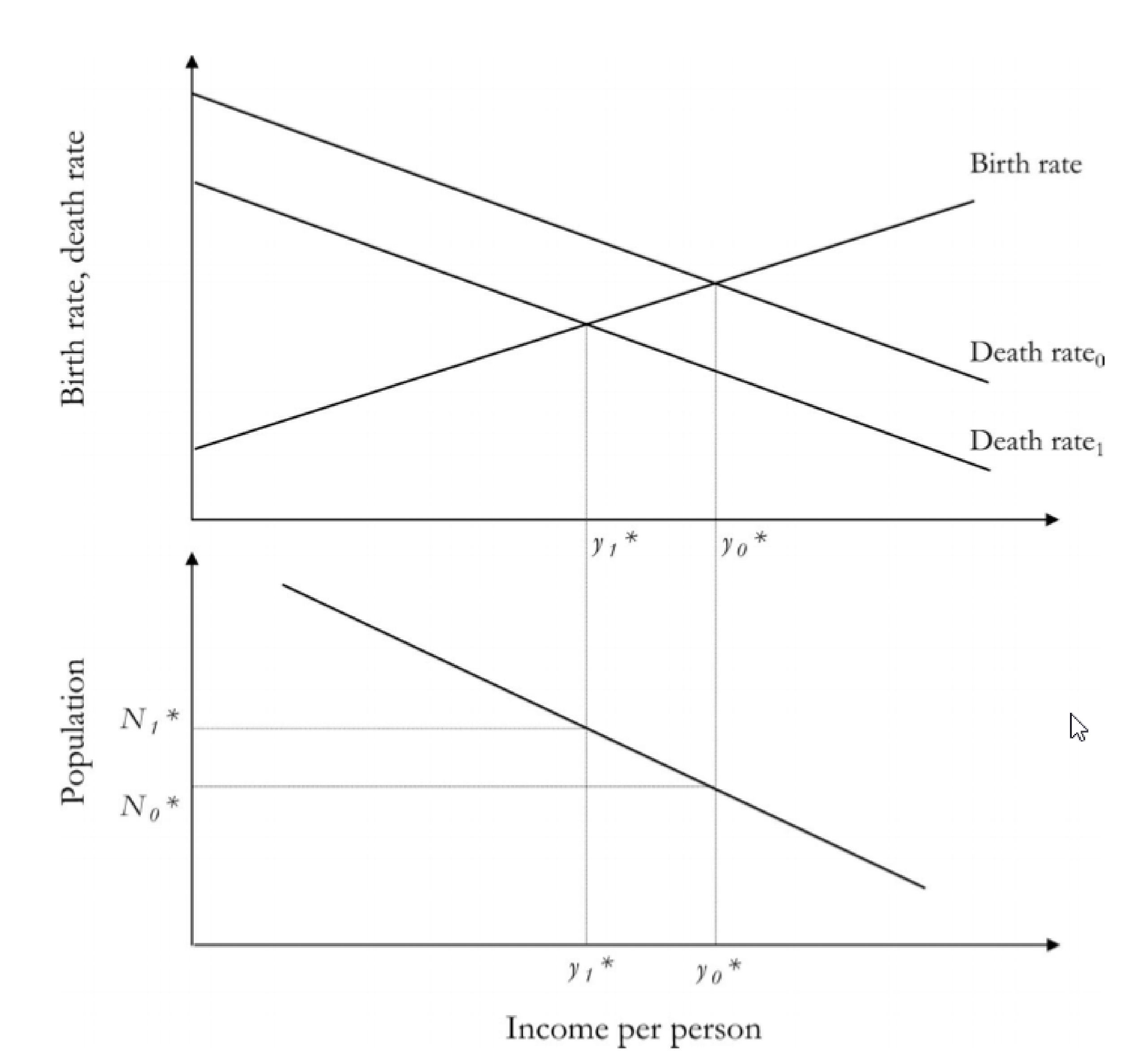
Why are incomes in Malawi today below those of England 1800

Greg Clark *Farewell to Alms*:

The English experience also shows that, while the Malthusian economy displayed stagnant material living standards, these were not necessarily low standards of living, even by the measure of many modern economies. Though the consumption pattern of the preindustrial English worker around 1800 may seem primitive, it actually implies, from the shares devoted to different goods, high living standards by the measure of the modern Third World. Over 40 percent of the food consumption, for example, was for luxury goods like meats, milk, cheese, butter, beer, sugar, and tea (see table 3.1). All of these are very expensive sources for the calories and proteins necessary to work and to maintaining the body. Very poor people do not buy such goods.

Effect of lowered death rate

- Modern medicine (e.g. antibiotics, vaccines) has lowered death rates in poor areas.
- In a purely Malthusian setting, this leads to *lower* per-capita income
- Death rate falls (curve shifts down)
  - At initial y_0^*, birth rate now exceeds new death rate
    - popn rises N_0^* to N_1^*
    - income falls y_0^* to new lower y_1^* steady state.
- (Fortunately, health campaigns + new income generation often trigger



European Marriage and Fertility

Table 4.2 Age of Marriage of Women and Marital Fertility in Europe before 1790

Country or group	Mean age at first marriage	Births per married women	Percentage never married	Total fertility rate
Belgium[a]	24.9	6.8	—	6.2*
France[a,b]	25.3	6.5	10	5.8
Germany[a]	26.6	5.6	—	5.1*
England[a]	25.2	5.4	12	4.9
Netherlands[c]	26.5	5.4*	—	4.9*
Scandinavia[a]	26.1	5.1	14	4.5

Sources: [a]Flinn, 1981, 84. [b]Weir, 1984, 33–34. [c]De Vries, 1985, 665.

Note: * denotes values inferred assuming missing values at European average.

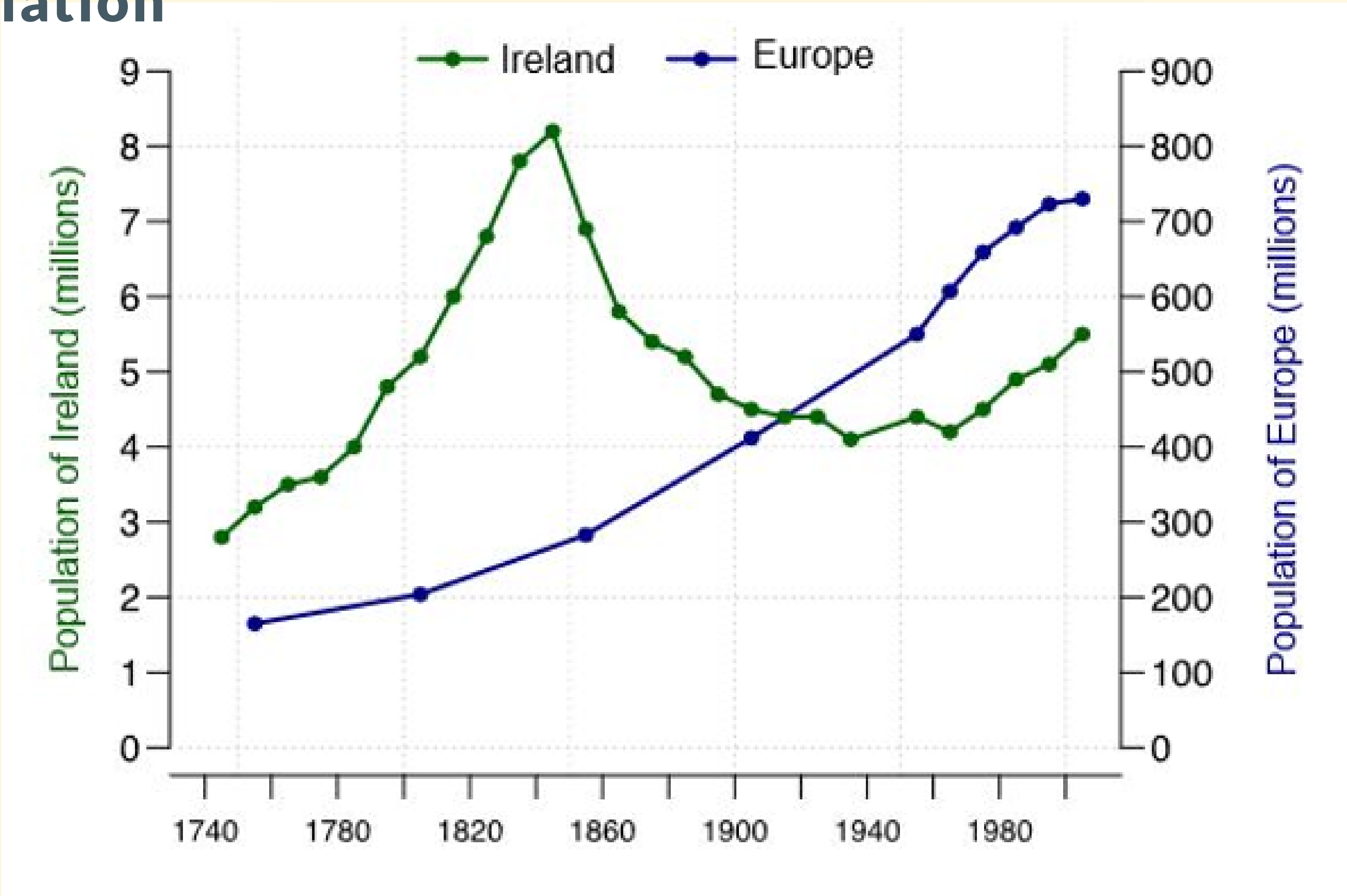
Why would modern day Malawi incomes be below English of 1800?

- Lots of explanations to be explored
- A simple Malthusian one:
  - modern medicine has lowered death rates but birth rates remain high.
  - $\Rightarrow$ higher population and lower incomes.

1. The Potato and the Irish Population

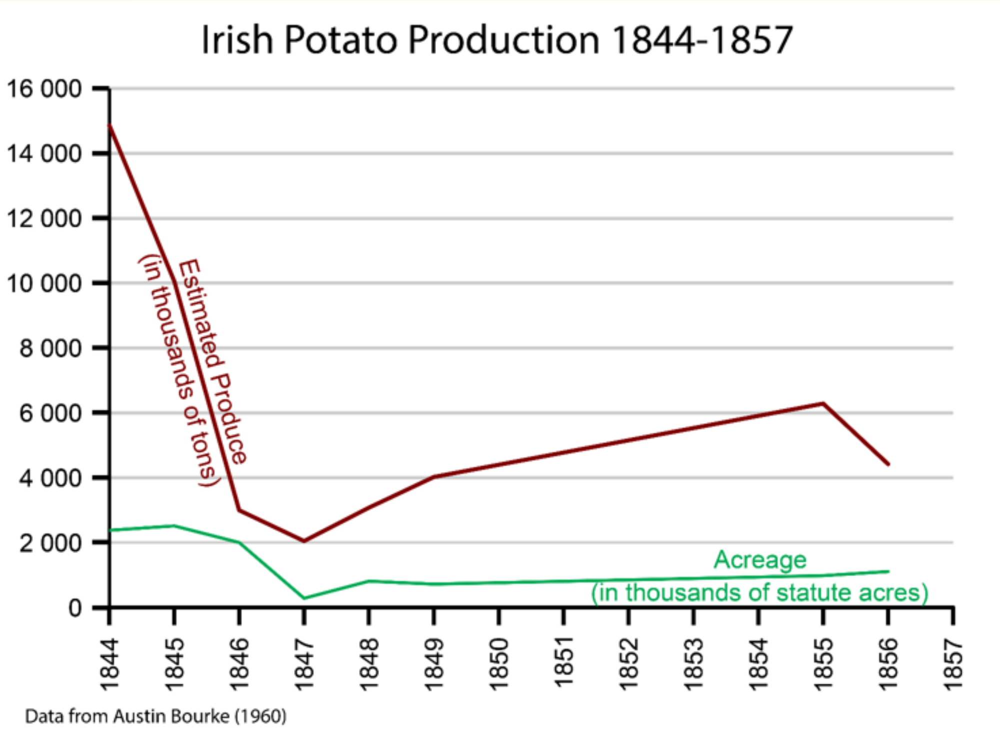
- Potatoes yielded 2–4x more calories per acre than grain; thrived in poor, damp Irish soil.
- Demographic Boom: Population doubled (4M to 8M+) between 1750–1845 as small plots could now support large families.
- The Lumper Monoculture: Dependence on a single, high-yield variety created a "genetic bottleneck" with zero disease resistance.
- The Blight: Phytophthora infestans arrived in 1845; the lack of genetic diversity caused total, rapid crop failure.

Irish population



1. Pre-Famine Context and The Blight (1845–1848)

- **The Potato Trap:** High-calorie yields fueled a population surge to 8M+, but dangerous "Lumper" monoculture.
- **Colonial Tenancy:** A rigid land system left the Catholic majority as landless tenants-at-will, dependent on tiny plots.
- **The Blight:** *Phytophthora infestans* arrives in 1845, causing total collapse of the primary food source.
- **Productivity Crash:** Decimation of the potato crop destroyed real wages of the rural poor, causing immediate destitution.



2. The Great Famine

- **Entitlement Failure:** Starvation occurred amid plenty; Ireland remained a net **exporter** of grain and livestock to England.
- **Ideological Failure:** *Laissez-faire* policy and "Poor Laws" prioritized market mechanics and property rights over direct relief.
- **The "Clearing":** Rent defaults triggered mass evictions, allowing landlords to consolidate small plots into large-scale cattle grazing.
- **Demographic Scarring:** 1 million dead; 1 million+ emigrated. Population has never returned to its 1845 peak.
- **Political Aftermath:** Perceived British indifference fueled century-long resentment and Irish nationalist movements.

3. Malthusian Classical Orthodoxy

- **The Malthusian Trap:** Famine was framed as a "natural check" on a population that had geometrically outstripped its food supply.
- **The "Iron Law":** Based on the *Iron Law of Wages*, officials feared that providing food would artificially lower mortality, leading to an even larger population collapse later.
- **New Poor Law (1834):** The Malthusian framework mandated that relief be "less eligible" (worse) than the lowest paid labor, forcing the starving into punitive workhouses.
- **Moralism:** High-ranking officials like Charles Trevelyan viewed the blight as a "divine mechanism" to forcibly modernize Irish social and property structures.

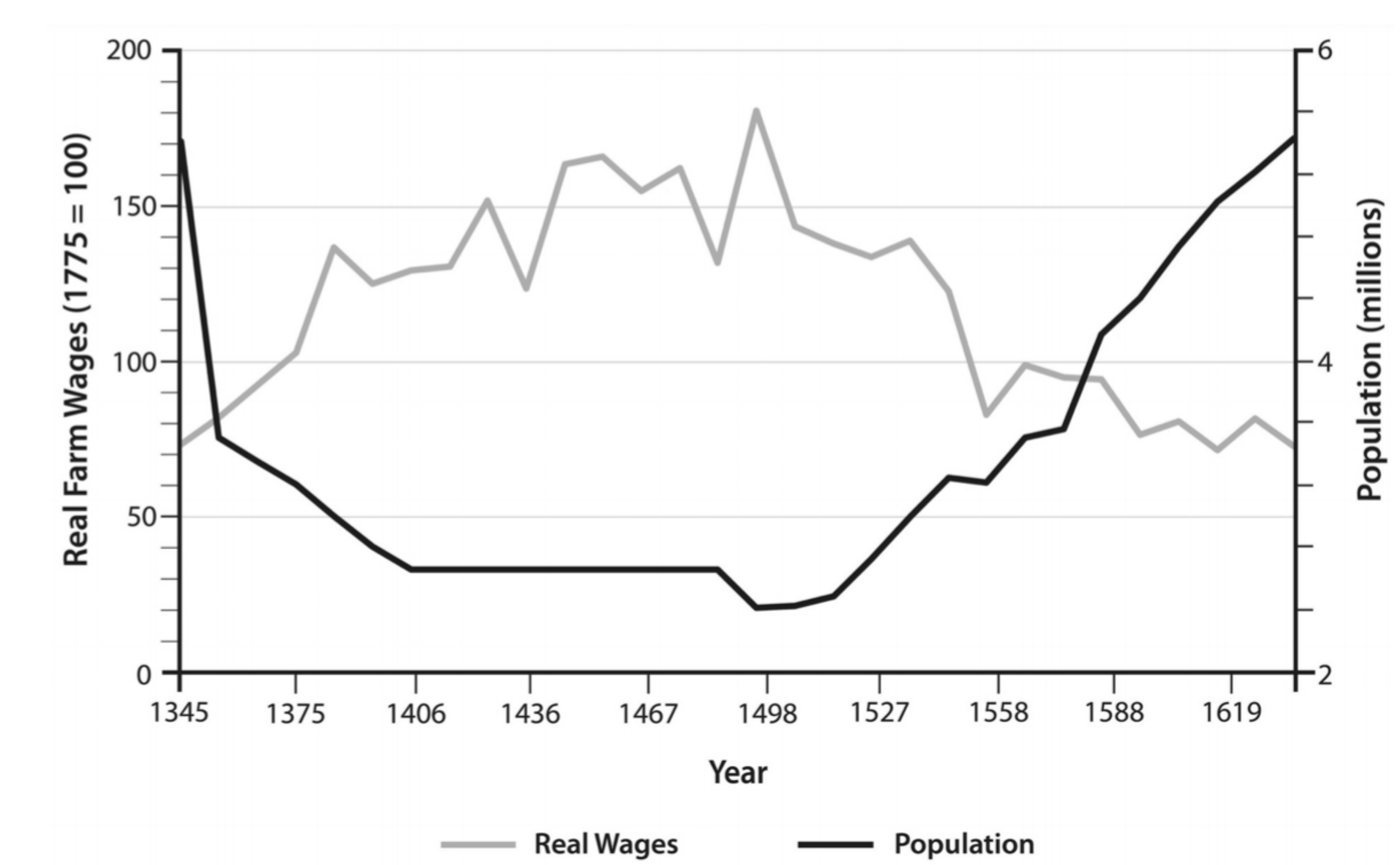
The Black Death

Table 5.1 Black Death mortality by country

Country	1300 population (millions)	Mortaliy estimate (%)
Austria, Czech Republic, & Hungary	10.0	20
Belgium	1.4	22.5
England	6.0	55
France	16.0	50–60
Germany	13.0	20–25
Italy	12.5	50–60
Netherlands	0.8	30–35
Poland	2.0	25
Scandinavia	1.9	50–60
Spain	5.5	50

Source: Jedwab, Johnson, and Koyama (2019). Original data from Ziegler (1969), Gottfried (1983), and Benedictow (2005).

The Black Death



Institutional change following the Black Death

- Elites attempted 'Seigneurial Reaction' to limit rise in real wages. But ultimately failed.
    - The Stature of Laborers, 1349 .
    - Serfdom dissappears in most West Europe
    - Emerged/hardened in parts of East Europe were landlords more powerful.
    - Voigtlander & Voth (2013) (slides below) argue Black Death led to new eqn:
        - Higher death rate
        - More urbanization
        - HIgher per-capita incomes

Black Death mortality rates (%) in 1347–52
Source: Jedwab, Johnson, and Koyama (2019)

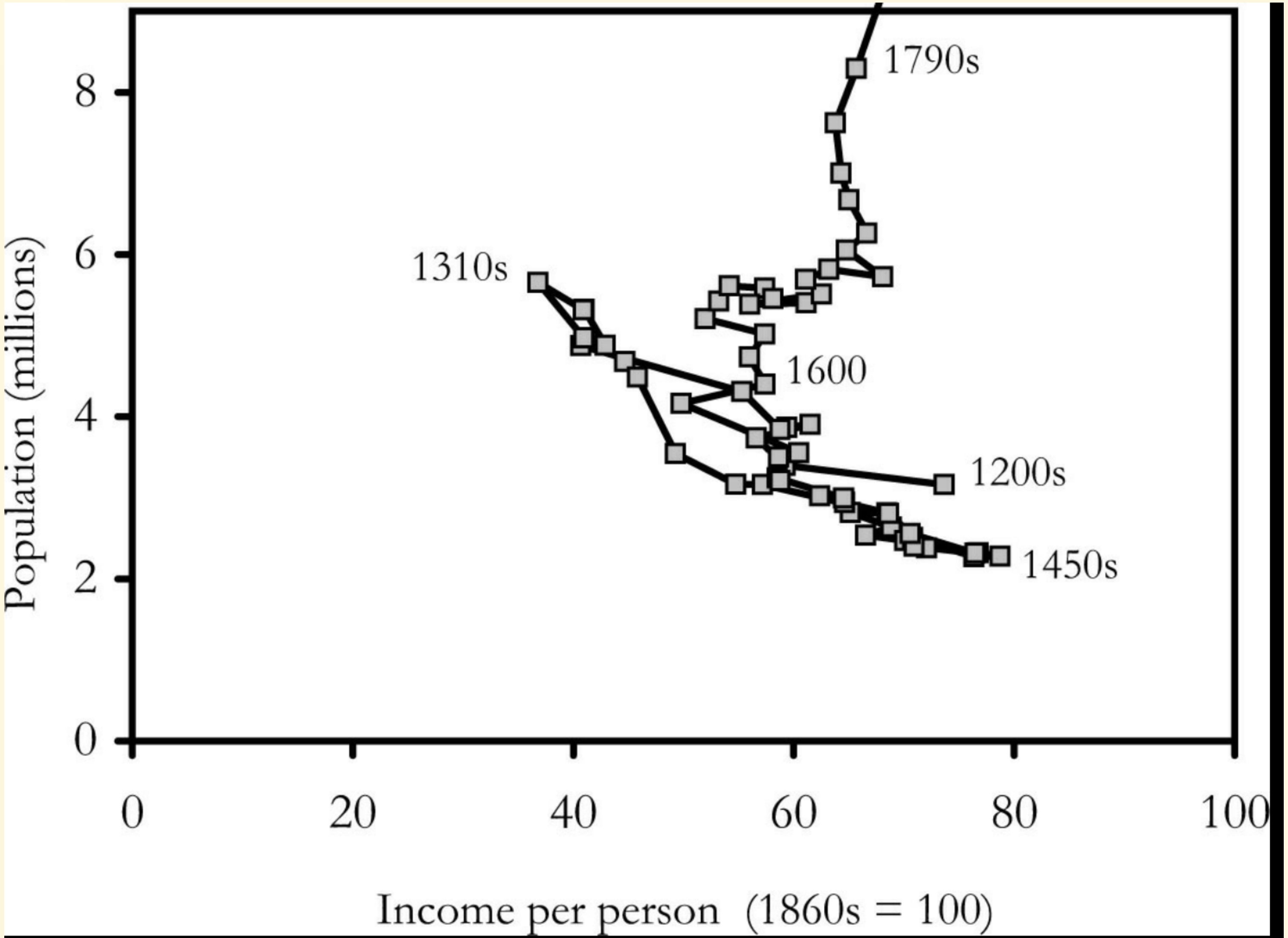


Figure 2.6 Revealed technological advance in England, 1200–1800.

- English data 1200-1790s
- Note clear effects of the Black Death > 1340s
- Transition ~ 17th century
    - Population rising without falling income per capita
- Sustained rising income per capita > 1870s

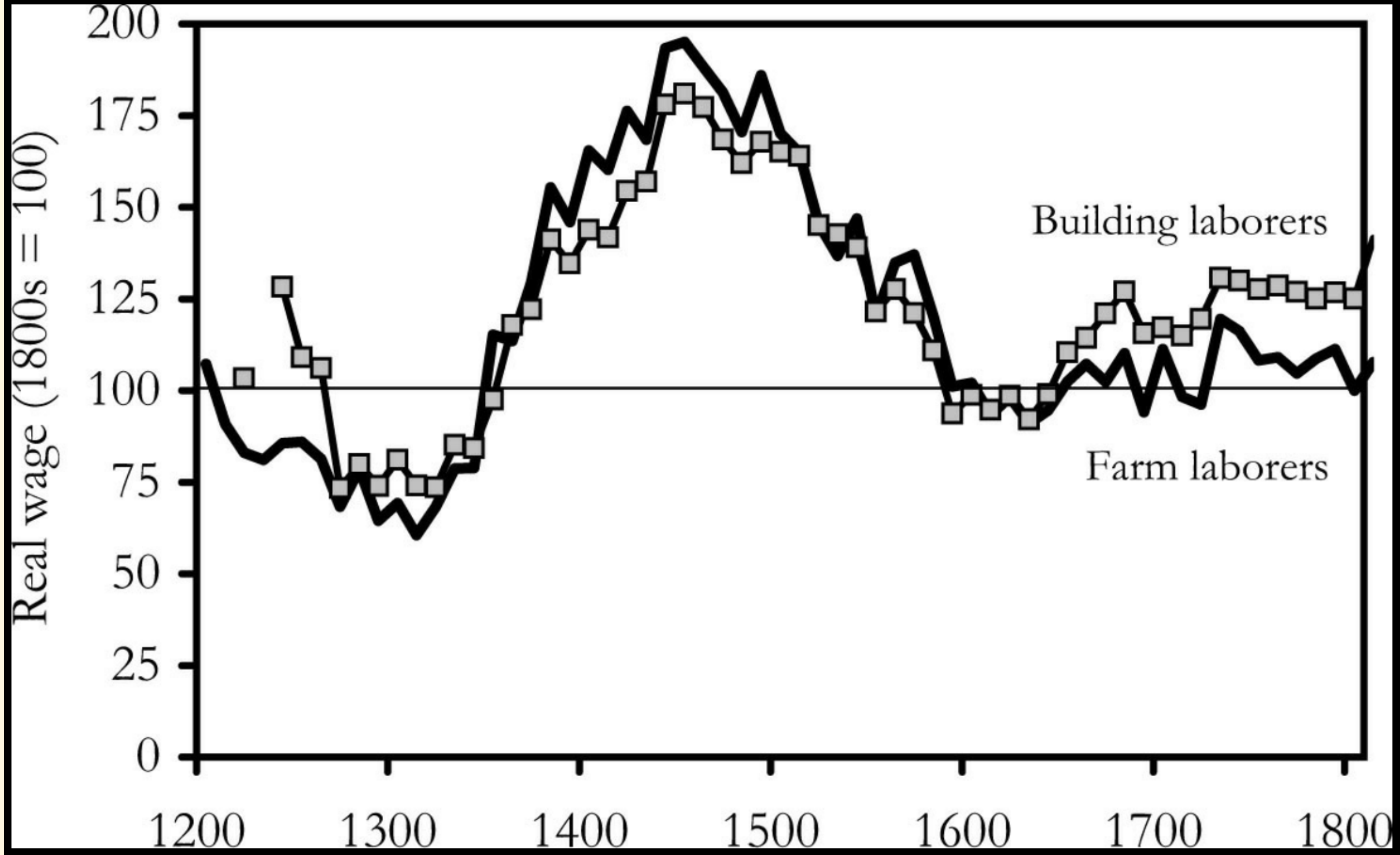


Figure 3.1 English laborers' real wages, 1209–1809.

Did plagues lead Europe to transition?

The Three Horsemen of Riches: Plague, War, and Urbanization in Early Modern Europe

NICO VOIGTLÄNDER
UCLA Anderson School of Management and NBER

and

HANS-JOACHIM VOTH
ICREA CREI, Barcelona GSE, and Department of Economics, Universitat Pompeu Fabra

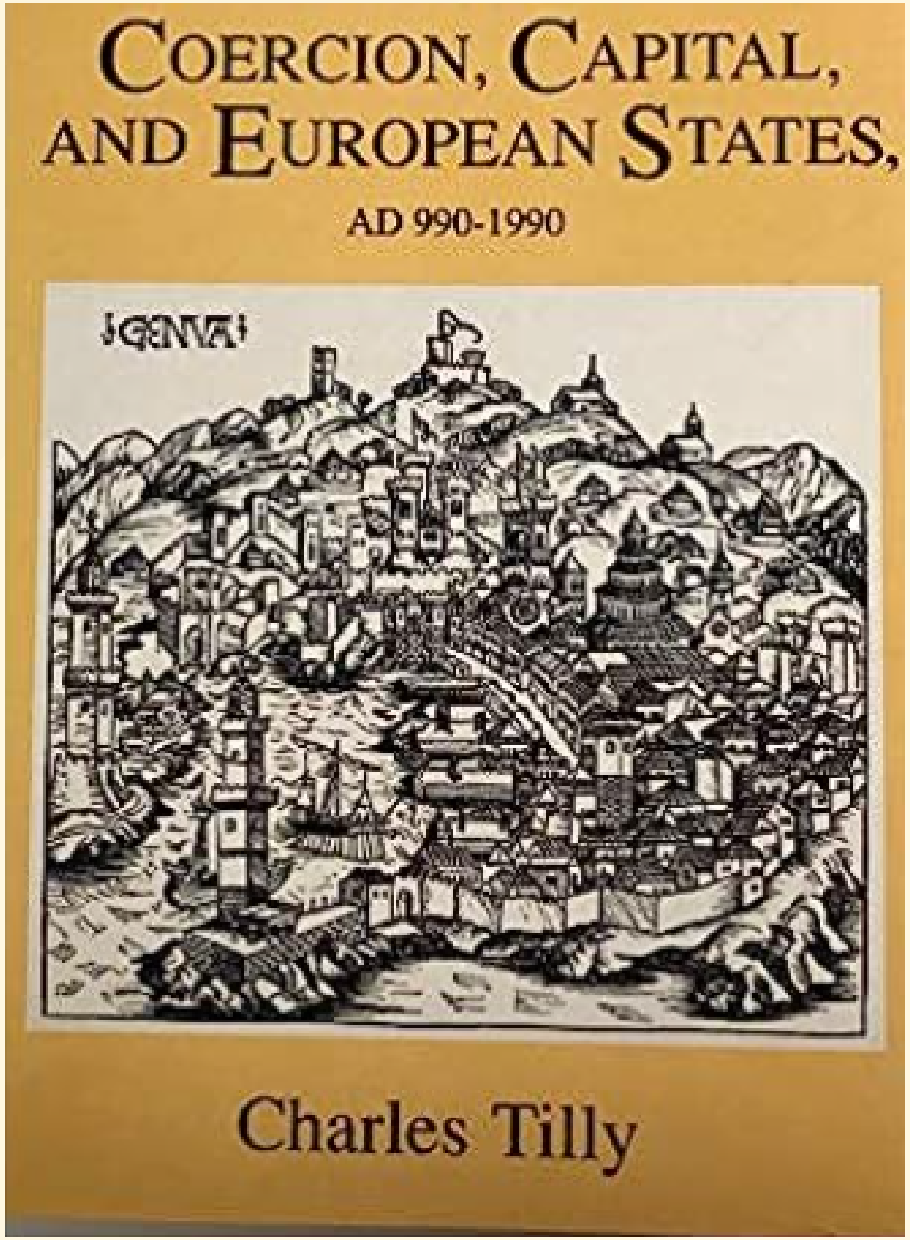
Did population shocks move the economy to a better steady state?

- Paper title play on *Four Horsemen of the Apocalypse* (symbolic representations of the end of times in Christian scriptures): Conquest/Pestilence, War, Famine, Death.
- Argument:
  - Normally death rate curve falls with income per capita y
  - But during 1350-1700 period **death rate curve at times rose with** y
    - **Plague** $\rightarrow$ per-capita income $y \uparrow$
    - $y \uparrow \rightarrow$ **Urbanization** $\uparrow$ (e.g. rise manuf demand and trade)
      - mortality was higher in cities so urbanization drove up death rate
    - $y \uparrow \rightarrow$ **War** $\uparrow$ (e.g. more tax for wars), death rates up.
- Possible multiple-equilibria, a shock may move economy from low to high equilibrium.

Wars and State Formation

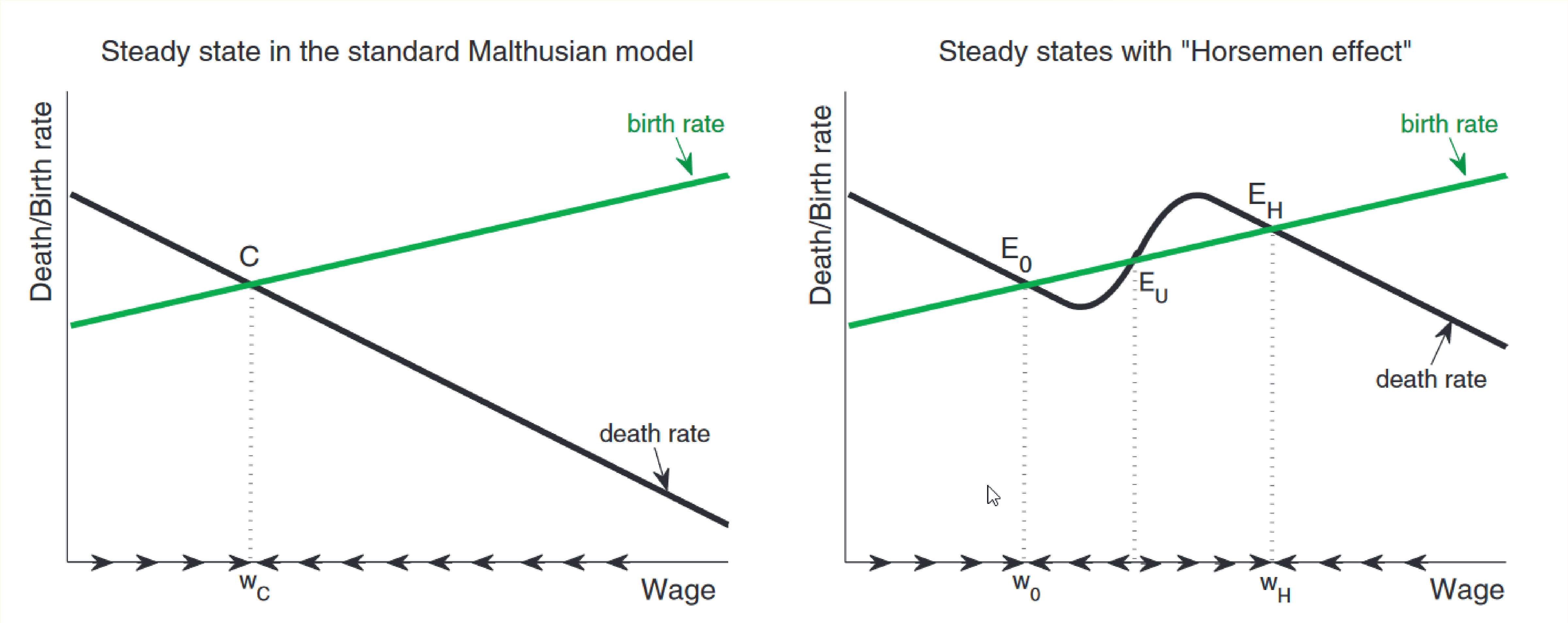
Sociologist **Charles Tilly** estimates that over 1500-1800, European Great Powers were engaged in wars 9 of every 10 years.

- To succeed/survive, states compelled to develop more centralized and sophisticated organization
- Strong centralized states with modern bureaucratic administration emerged.

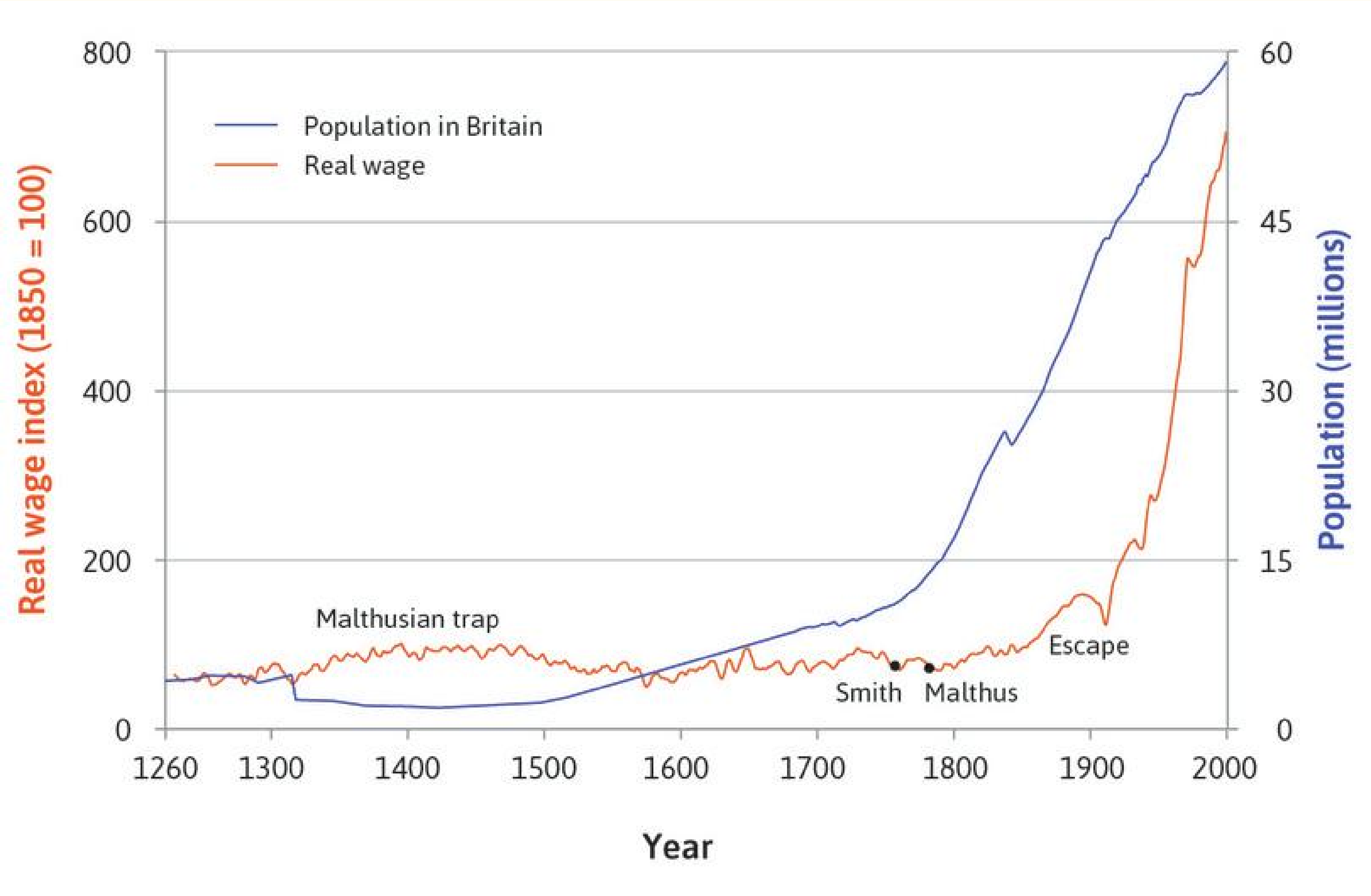


How did Europe escape the "Iron Law of Wages?" We construct a simple Malthusian model with two sectors and multiple steady states, and use it to explain why European per capita incomes and urbanization rates increased during the period 1350–1700. Productivity growth can only explain a small fraction of the rise in output per capita. Population dynamics—changes of the birth and death schedules—were far more important determinants of steady states. We show how a major shock to population can trigger a transition to a new steady state with higher per-capita income. The Black Death was such a shock, raising wages substantially. Because of Engel's Law, demand for urban products increased, and urban centers grew in size. European cities were unhealthy, and rising urbanization pushed up aggregate death rates. This effect was reinforced by diseases spread through war, financed by higher tax revenues. In addition, rising trade also spread diseases. In this way higher wages themselves reduced population pressure. We show in a calibration exercise that our model can account for the sustained rise in European urbanization as well as permanently higher per capita incomes in 1700, without technological change. Wars contributed importantly to the "Rise of Europe", even if they had negative short-run effects. We thus trace Europe's precocious rise to economic riches to interactions of the plague shock with the belligerent political environment and the nature of cities.

Shock to populaiton

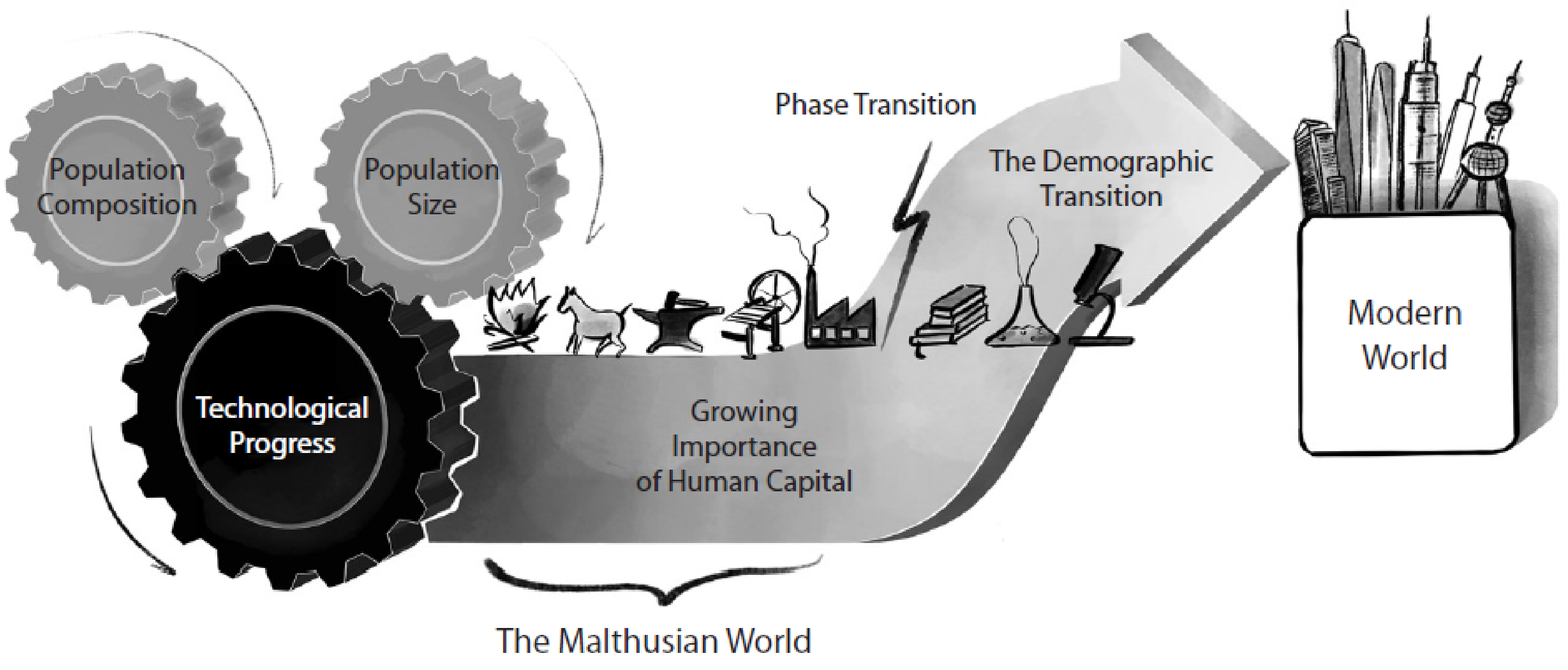


Escape



Population in Britain
Real wage
Real wage index (1850 = 100)
Population (millions)
800
600
400
200
0
60
45
30
15
0
Malthusian trap
Smith
Malthus
Escape
1260
1300
1400
1500
1600
1700
1800
1900
2000
Year

The Storm Beneath the Surface



The Cogs of Change

Quantys versus Qualys

- Quantys: 'be fruitful and multiply'
  - 4 children per family, 2 survive to adulthood
  - little investment in human capital
  - subsistence farmers, manual laborers, etc.
- Qualys: 2 children per family
  - 2 survive to adulthood
  - heavier time and resource investment in child productivity/earning capacity
  - blacksmiths, traders, carpenters.
- Both groups maintaining shares in the population

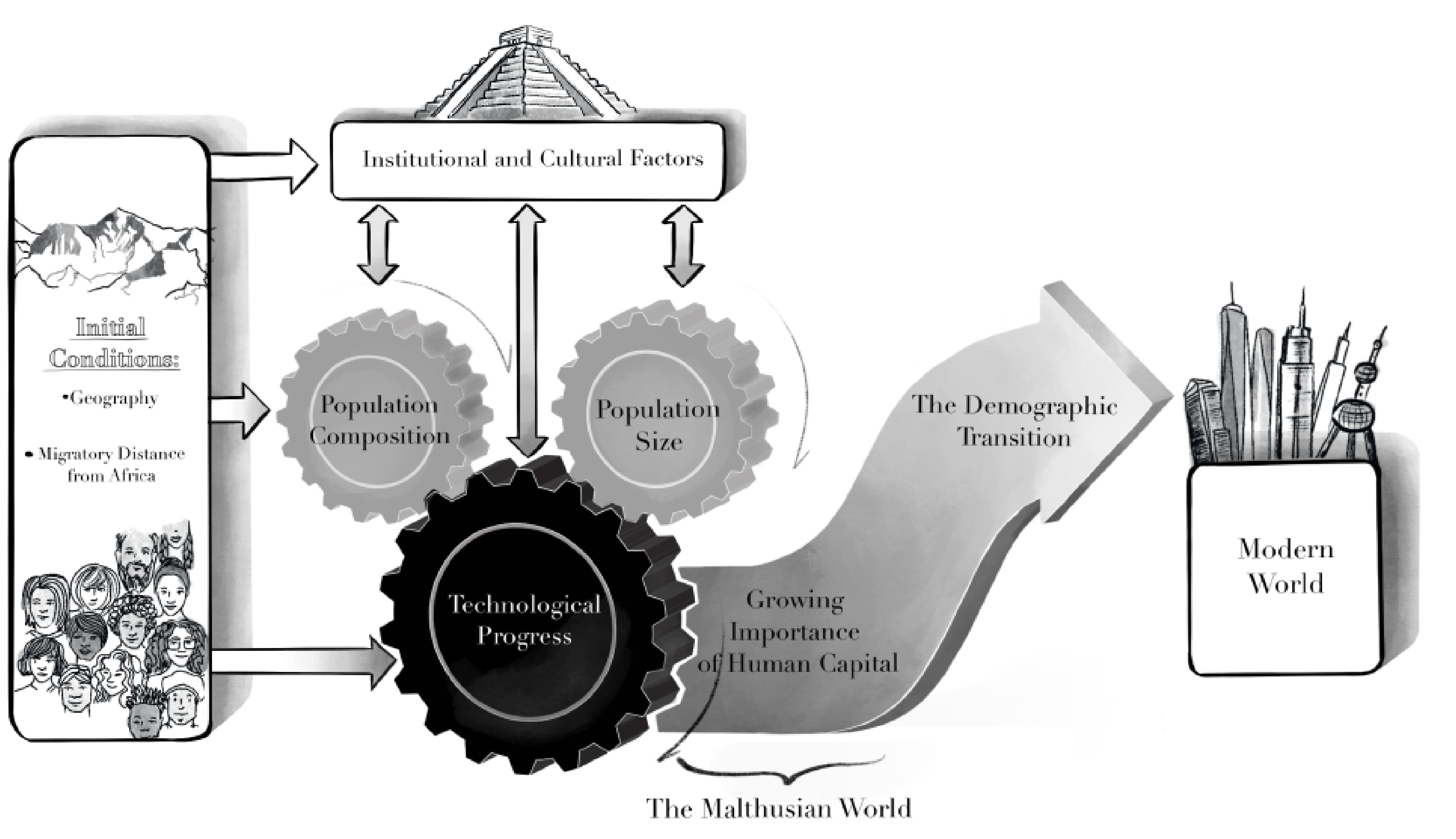
Quantys versus Qualys

Suppose technological development boosts demand for qualy services

- raises income or Qualys relative to Quantys (an evolutionary advantage)
- Over time raise family size from 2 to 3 children
- Still below rate of Quantys, but now more Qualy children reach maturity

Positive feedback loop

- tech $\rightarrow$ Qualy incomes and population $\rightarrow$ more tech.


Institutional and Cultural Factors
Initial Conditions:
•Geography
• Migratory Distance from Africa
Population Composition
Population Size
Technological Progress
Growing Importance of Human Capital
The Demographic Transition
Modern World
The Malthusian World

Elements of Galor's Unified Theory

Population size and Composition foster progress via:

- supply of innovations
- Demand for innovations
- Diffusion of Knowledge
- Division of Labor
- Extent of Trade

Traits complementary to the growth process (transmitted culturally as norms/customs)

- Generated higher incomes
  - capital accumulation
  - future-orientation
- Adaptation

A simpler model

From Malthus to Solow

- **Malthusian growth model:** Land and labor.
  - population growth is exponential, land is fixed (non-accumulated).
  - One time improvement in technology $A \to$
    - income per capita y falls back toward subsistence.
    - long run growth in per-capita incomes essentially zero.

Example:

$$Y = A \cdot T^{\beta} \cdot L^{1-\beta}$$

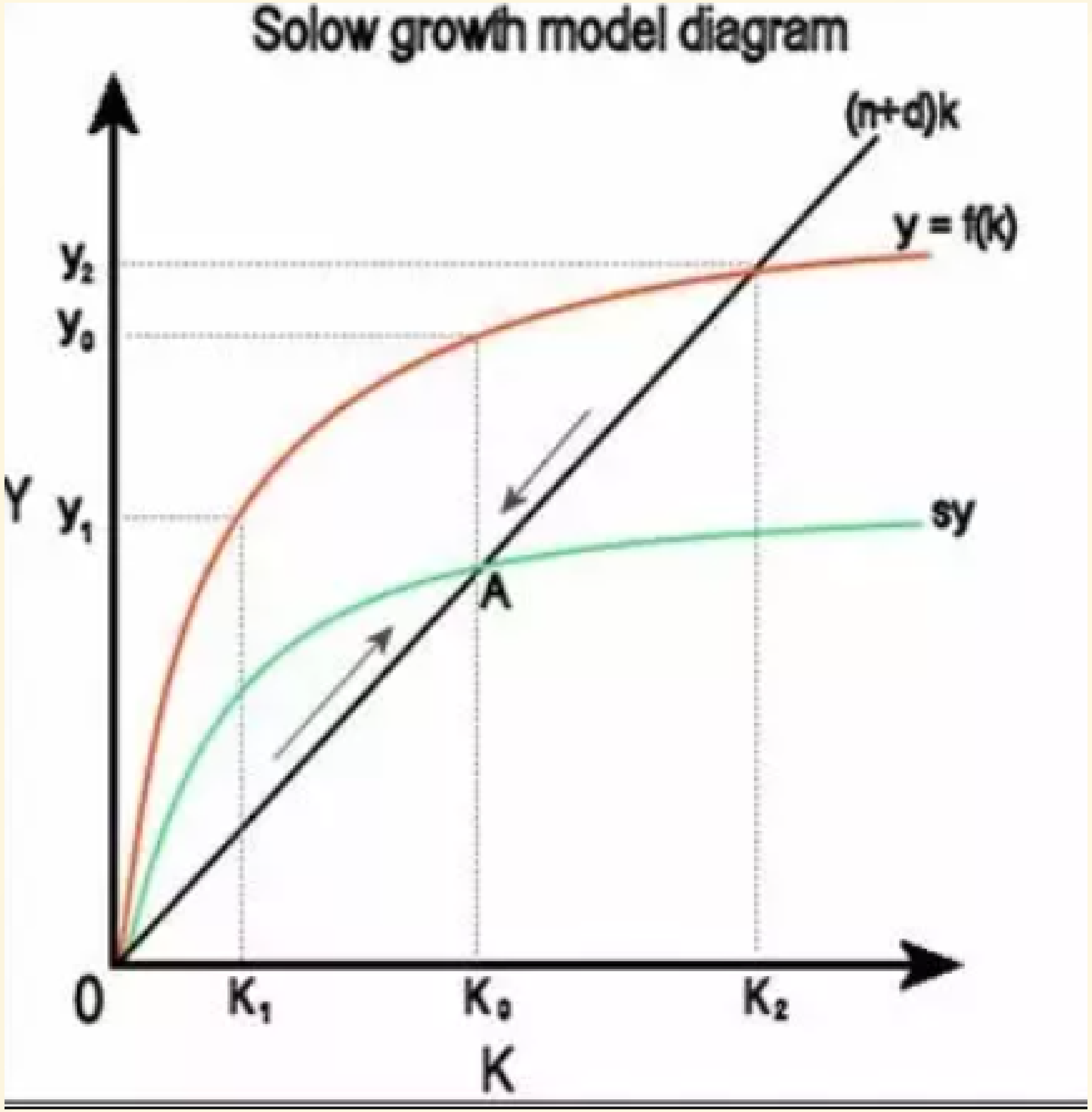
but land T is fixed, so $y = A \cdot \frac{1}{L^{\beta}}$

- Higher $A \to y$ rises temporarily, but eventually falls back to subsistence level.

Solow Growth Model

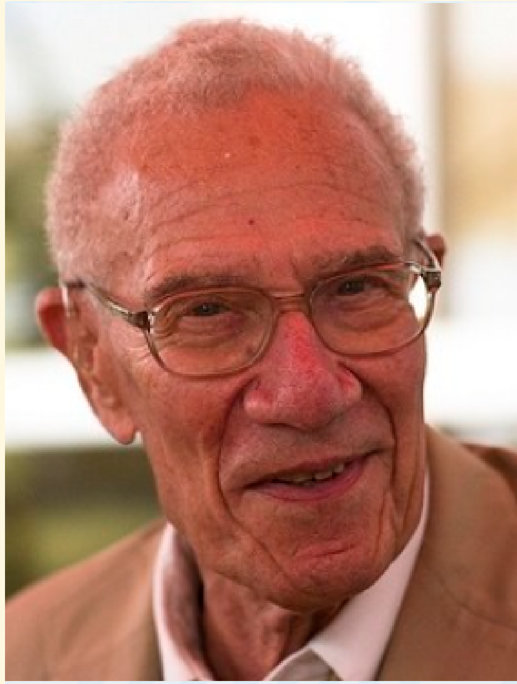
The early main "neo-classical growth model":

- Production with labor and capital (could add human capital).
- Unlike land, capital can be accumulated via saving/investment.
- New capital tools to equip expanding worker population.
- But capital accumulates subject to diminishing returns.
- In steady-state
  - Capital per worker will be constant $k = \frac{K}{L}$
  - Income-per capita y constant at level above subsistence, despite pop growth.
  - Investment make up for capital depreciation and equip new workers.
  - Level determined by technology A, savings rate s, and depreciation rate δ.



Solow growth model diagram
(n+d)k
y = f(k)
sy
A
Y
y2
y0
y1
0
K1
K0
K2
K

- **Solow (neo-classical) growth model:** capital is accumulated.
    - growing population of workers can be equipped with new capital.
        - capital accumulation subject to diminishing returns.
        - economy reaches steady-state level y above subsistence.
    - One time improvement in technology $A \to y$ rises to higher steady state.
    - Long run *growth* in per-capita income only with *sustained* technology improvement.
- Robert Solow (MIT), Nobel 1987



> "In the 1950s, he developed a mathematical model illustrating how various factors can contribute to sustained national economic growth. From the 1960s on, Solow’s studies helped persuade governments to channel funds into technological research and development to spur economic growth."

Neo-classical growth model (expanded Solow model)

- Predicts 'convergence' of per capita income across countries.
  - Suppose two countries with same technology and same savings rate.
    - High capital/worker $k = K/L$ country is closer to steady-state. Grows slowly.
    - Country w/ low capital/worker $k = K/L$ further away to steady-state. Grows faster.
    - Model predicts **convergence** of per capita income over time.

Neo-classical growth more generally

- Convergence in per-capita incomes across countries, via any that tends to equalize technology and capital/worker ratios across countries. Market-based mehanisms will do the job:
  - **via capital accumulation** and/or international **capital flows**.
  - via **labor flows** across countries
  - via comparative advantage and **trade**.
- Premised on no market failures and diminishing returns to capital.
- Development policy implications:
  - Let markets do their job: Liberalize, privatize, deregulate.
  - Poor countries are poor because of distortions to markets.